

Social robot navigation based on HRI non-verbal communication

A case study on avocado harvesting

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ABSTRACT

To date, robotic applications in agriculture are still a challenging topic, which has been studied mainly for large farms. However, groves in particular, require tasks, such as picking and handling, that still require human labor force. In countries such as Chile and Peru, avocado is one of the main fruit production, but its growing in complex environments, making it difficult to fully automate the harvesting process. In this scenario, human-robot interaction (HRI) strategies can provide solutions to enhance the farming process. In this work, we propose the use of a HRI strategy via three visual non-verbal communication methods, with the aim of improving the avocado harvesting process leading to possible human workload decrement. Using such HRI directives, a robot motion controller is implemented for the robotic service unit to ensure that the interaction is socially acceptable during the avocado transportation task. The robot social navigation is tested in a simulated environment where the robot interacts with field workers to test three control tasks which are approaching, following and avoiding the human.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design process and methods**; *Collaborative interaction*; • **Computing methodologies** → **Planning for deterministic actions**; **Vision for robotics**; *Activity recognition and understanding*;

KEYWORDS

Social robot navigation, Human-robot interaction, non-verbal communication, avocado harvesting

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1 INTRODUCTION

Nowadays, agricultural processes are commonly carried out by humans, operated machines and different autonomous mobile robots that usually work in large scale fields. However, in several countries, harvesting processes have been done mainly by hand and using human-operated machinery, since there are still processes too complex to automate completely. In this context, a human-robot interaction (HRI) strategy can provide several solutions to help farmers and improve the avocado harvesting process. The HRI field is aimed at studying, designing, understanding and evaluating the interaction between humans and robots, in order to reach a natural and fluid cooperation [4, 14]. The interaction between humans and robots can allow certain tasks to be done while sharing physical workspace, performing an intuitive and easy communication with the robot through speech, non-verbal gestures, among others [12]. In the case of agriculture, several HRI approaches have been studied from different perspectives such as interface designs [1], interaction methods [2], collaboration models [11], among others. In particular, the advantages seems to be relevant in activities such as crop detection, spraying, grasping, detaching, and transporting procedures [1]. However, strategies and architectures that allow humans and robots to work at field cooperatively, in a natural manner, without using an interface, and sharing the workspace, are still an open research topic. An appropriate HRI strategy should permit the farmer to communicate with the robot in an intuitive manner during harvesting, while both parts work by sharing the workspace in a socially accepted manner to optimize the harvesting process.

2 HRI FOR AVOCADO HARVESTING PROCESS

In this work, a HRI architecture for the avocado harvesting process is presented in Fig. 1. It can be observed that the human farmer can perform harvesting from ground or using a ladder, and then he/she interact with the robot to deposit the collected crops on it. The robot mission is to transport the crop boxes to the cargo vehicle zone, where a human receive them, perform a visual inspection, and deposit them on the cargo vehicle. The robot is always navigating among the field using socially accepted criteria, until a human require its assistance. For detect human intention, three human-robot non-verbal communication methods are proposed. The HRI architecture illustrated aims at helping to reduce human workload and increase the efficiency of the process by increasing the crop volume harvested per day.

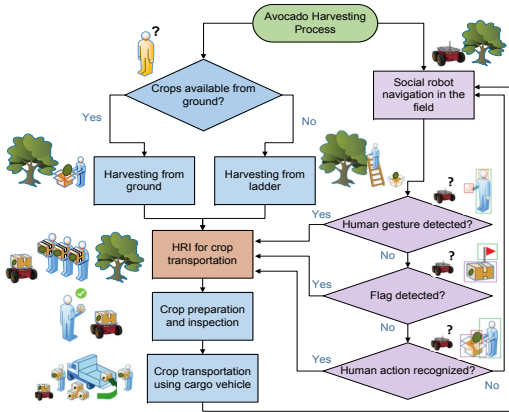


Figure 1: HRI architecture for avocado harvesting process.

3 DETECTING HRI NON-VERBAL COMMUNICATION FEATURES

In the present work, three different HRI methods are analyzed in which humans and robots can interact communicating through the use of non-verbal techniques during avocado harvesting process. These methods are based on convolutional object detection approaches, as is explained in section 3.1. The robot directives are based on the HRI methods, which are: human hand gestures detection (section 3.2), marking flags detection (section 3.3), and human action recognition (section 3.4).

3.1 Convolutional object detection approaches

In this work, TensorFlow object detection application programming interface (API) [7, 8] is used to detect hand gestures and marking flags during robot navigation in an avocado harvesting scenario, as showed in Sections 3.2 and 3.3. The convolutional trained model for these sections is Faster R-CNN as the meta-architecture, and Inception v2 as the feature extractor. Detailed information about architecture, hyper-parameter tuning, and model features can be found in [7, 8]. The training-set used contains 3004 images of 5 subjects, and 227 to test the algorithm performance. The images were taken in the avocado farms in the Regional Research Center (INIA), La Cruz, Chile. The illumination levels were between 1316 in the shade and 54900 luxes daylight. The image size used was 1280x720 pixels. The algorithm can detect nine object classes, which are: human, face, box, flag and five different hand gestures. Each algorithm iteration takes 0.204 seconds using a computer with Nvidia Geforce GTX 970M video card processor and 32Gb RAM, which allow to make decisions on-line during vehicle navigation.

The object detection results for each of the nine detected classes is illustrated in Table 1. For each one of the 227 images, several algorithm performance indicators were computed, which are: true positives (TP), false negatives (FN), false positives (FP), recall or true positive rate (TPR), precision or positive predictive value (PPV), F1 score (F1) and average precision (AP).

Table 1: Object Detection Results

No.	Class	TP	FN	FP	TPR	PPV	F1	AP
1	human	210	0	9	100%	96%	98%	99%
2	face	148	25	1	86%	99%	92%	91%
3	box	165	16	0	91%	100%	95%	98%
4	flag	42	3	6	93%	88%	90%	95%
5	gesture_1	29	3	4	91%	88%	89%	82%
6	gesture_2	23	0	3	100%	88%	99%	93%
7	gesture_3	34	1	0	97%	100%	99%	93%
8	gesture_4	85	16	1	84%	99%	91%	85%
9	gesture_5	12	0	17	100%	41%	59%	81%

3.2 Detecting hand gestures

Non-verbal communication problems such as hand gesture, eye gaze or emotion recognition can provide valuable information that allow humans to interact with robots in an intuitive manner for certain tasks [12]. The robot, which will be navigating among the field rows, can detect a human and evaluate if he/she requires assistance based on the hand gestures detected. The actions that a robot can perform based on such gestures are: reach human position (gesture_1), return to cargo vehicle (gesture_2), follow the human (gesture_3), stop current task (gesture_4), and continue previous task (gesture_5) as illustrated in Fig. 2.a.

3.3 Detecting marking flags

Once the human finishes filling the box of avocados during the harvest, he can raise a visual flag on the box and continue the crop collection task. The robot, which will be navigating along the farm, can approach to the position of the box if it detects the flag raised as shown in Fig. 2.b, and wait until the human places the box on its back and make the gesture to return to the cargo vehicle.

3.4 Human action recognition

In [15], we shown that it is possible to use semantic representations based on object detection to estimate human activities in still images from the avocado harvesting processes (see also [5, 16]). This information can be used to detect when a human is performing a crop collection process from the ground or from a ladder as illustrated in Fig. 2.c. The robot should only approach the farmers who perform such tasks, wait until the farmer finishes filling the box with avocados, then receive the box and then return to the cargo vehicle zone.

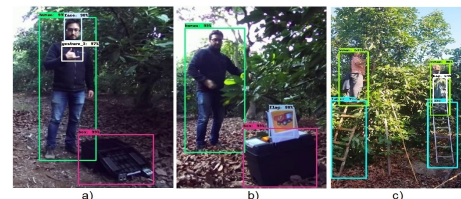


Figure 2: a) gesture detection b) flag detection c) human action recognition.

4 HRI FOR ROBOT MOTION CONTROL

In order to reach collaboration among the human and robot during the harvesting process, it is necessary to consider an appropriate HRI control strategy for the robot, which is illustrated in Fig. 3. The decision making stage depend of non-verbal communication methods explained in section 3. Such methods provides the HRI directives to the robot social motion planning, which generate a reference guidance trajectory and determine maneuverability constraints to obtain a safe and sociable acceptable robot motion.

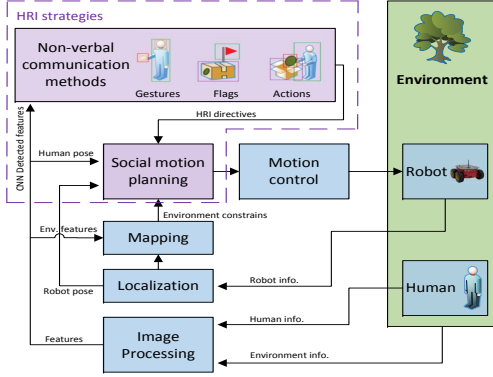


Figure 3: HRI control strategy for mobile robot system

4.1 Social Navigation

In this work, three HRI directives are studied: robot avoiding humans, robot approaching to a human and robot following a human. Such control tasks are address using concepts such as Human-robot shared workspaces [13], social navigation [3, 10] and social field avoidance [6].

To obtain a safety and sociable acceptable robot approaching to a human when he/she requires, it is necessary to consider interaction parameters such as safe distances and social acceptable orientations.

In Fig. 4 (left), different circular zones that conform human personal space are illustrated. According to [6], when a robot navigates by sharing the workspace with the human, it is necessary that it respects social zones to improve its social acceptance. Therefore, when a robot approach to the human, the distance d between the human and the robot should be within the social zone. Moreover, It is necessary to consider socially acceptable areas based on human vision range as shown in Fig. 4 (right), which illustrates several possible regions through which the robot can approach or follow (depending on the task) the human[3]. In this work, we consider that the robot is in front of the human if his/her face is detected, which is a desirable condition that the robot should accomplish if possible.

4.2 Motion Control

Considering only the kinematics of unicycle mobile robot, its states can be represented by $q_r = [\theta_r \ x_r \ y_r]^T$, that consist in orientation angle θ_r and the position (x_r, y_r) . Then, let us consider a typical trajectory tracking task with a reference guidance trajectory $q_d(t) = [\theta_d(t) \ x_d(t) \ y_d(t)]^T$, where $\theta_d(t)$ and $(x_d(t), y_d(t))$ are the desired

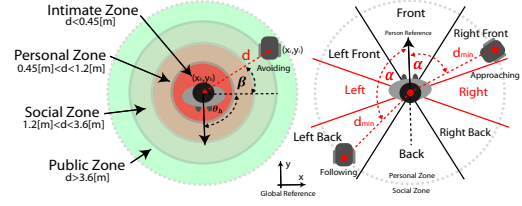


Figure 4: Social navigation considerations.

orientation and position respectively for any time instant t . The robot is controlled in function of error vector $e = q_d - q_r$ by modifying angular and linear velocities using the control law $\Phi(e)$, which is a simplification of the Vector-Field-Orientation (VFO) feedback control presented in [9]:

$$\Phi(e) = \begin{bmatrix} \Phi_\omega \\ \Phi_v \end{bmatrix} = \begin{bmatrix} k_a(\theta_a - \theta_r) \\ h_x \cos \theta_r + h_y \sin \theta_r \end{bmatrix} \quad (1)$$

where $\theta_a = \text{Atan2}(h_y, h_x) \in (-\pi, \pi)$, $h_x = k_p(x_d - x_r)$, $h_y = k_p(y_d - y_r)$ and k_a, k_p are design parameters to calibrate the controller.

4.3 Social Field Avoidance

Motivated by previous works [9, 13], a social field avoidance is included in the control law (1). The avoidance methodology is based on potential fields concept such that:

$$F(x, y) = \exp \left(-d^2 \left(\frac{(\cos(\beta - \theta_h))^2}{\sigma_x^2} + \frac{(\sin(\beta - \theta_h))^2}{\sigma_y^2} \right) \right) \quad (2)$$

where σ_x and σ_y are standard deviations used to deform the social field function $F(x, y)$ according to social space zones. The distance d and angle β (illustrated in Fig. 4) are calculated in based of the human pose $q_h = [\theta_h \ x_h \ y_h]^T$ according to:

$$d = \sqrt{(x_r - x_h)^2 + (y_r - y_h)^2} \quad (3)$$

$$\beta = \text{atan2}((y_r - y_h), (x_r - x_h)) \quad (4)$$

Finally, using the control law (1) as base, the so-called convergence vectors h_x and h_y are modified to add the evasion task in the following way:

$$h_x = k_p(x_d - x_r + F(x, y) \cos(\beta - \theta_h)) \quad (5)$$

$$h_y = k_p(y_d - y_r + F(x, y) \sin(\beta - \theta_h)) \quad (6)$$

4.4 Robot approaching and following a human

Human farmer following could be treated as a tracking task with proper criteria to recalculated a new reference guidance trajectory [10] such as:

$$x_d(t) = x_h(t) - (d_{min} \cos(\alpha + \theta_h(t))) \quad (7)$$

$$y_d(t) = y_h(t) - (d_{min} \sin(\alpha + \theta_h(t))) \quad (8)$$

where, $|\alpha| < \pi/3$ [rad] and d_{min} is a tuning parameter chosen in a range of values 1.2 [m] $< d_{min} < 3.6$ [m] according to the social zone.

The robot approaching use the same criteria to select the reference point, but such point is not variable in time, so it can be

treated as a typical set-point task. This control task use (7) and (8) with $|\alpha| < 7\pi/6$ [rad].

4.5 Simulation Results

To test the HRI strategies for social navigation, the three main control directives was simulated in a V-REP virtual ambient. The motion planning and control were programed in Matlab, which connect with V-REP to exchange sensor information and control signals. The Fig. 5 presents results of an complete experiment where a robot interact with three humans. The route followed by the robot is divided by colors because of each human interaction tested different control directives.

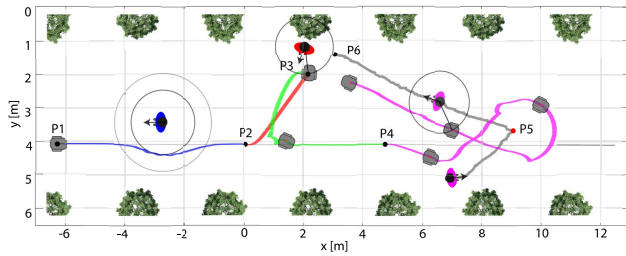


Figure 5: Path followed by mobile robot whilst it shared the work space with humans in avocado scenario.

The original reference trajectory (gray straight line) was generated from the robot initial position P1 with reference linear speed $v_d = 0.2$ [m/s]. The robot begins its navigation with a default trajectory tracking directive with social field avoidance. During this test section, the robot interacts with the first human (blue), evading him with a socially acceptable distance limited by $d_{min} = 0.9$ [m]. The robot navigates with this directive until point P2, where the second human (red) gives a new approach directive towards him with $\alpha = \pi/5$ [rad] (left side approaching). In this point, the robot updates the reference trajectory to a new set-point located in P3. When the robot is no longer required by the second human (red), the robot continues with the original straight line trajectory until point P4 where the third human (magenta) gives a new directive of following using techniques from Section 4.4. During this last test section the robot follow the human keeping an angle $\alpha = 4\pi/3$ [rad] (left side following). The path followed by the third human is colored in gray and show an L-shape path with a clockwise turn in the point P5 and finished in P6.

According to graphics results, the control strategy fulfills with the aim of maintain the distance d greater or equal than a preset safe distance d_{min} in all interactions. Furthermore, during approaching and following tasks, robot orientation respect to farmers is keeping in a range similar to preset α angle.

5 CONCLUSIONS

In this work, HRI strategies for avocado harvesting process are presented based on non-verbal communication methods which provide motion directives to a robot during social navigation. It is shown that is possible to build a HRI architecture based on the original harvesting process obtained from visualizations. The possibility of

humans sharing the workspace with robots open new opportunities to farmers to increase process efficiency and reduce human workload, which will be properly evaluated in future works. The presented HRI schemes aims to improve agricultural processes that may not be completely automated due to its complexity. This will be a key factor to address the lack of human labor and to increase food production to tackle the increasing population demands, which is fundamental to reach sustainable agriculture practices.

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